

Qode — Equity Backtesting Platform

Technical Documentation — Module Reference, API, Assumptions & Optional Features

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Live: <https://backtest-web.onrender.com>

Backend Modules

[app/data/universe.py](#)

Defines `UNIVERSE` — a list of 120 NSE-listed stock symbols across large, mid, and small cap segments covering sectors like banking, IT, pharma, FMCG, energy, and infrastructure.

[app/data/fetcher.py](#)

Fetches data from Yahoo Finance via `yfinance` :

- `fetch_prices(symbol, start, end)` — returns daily OHLCV DataFrame
- `fetch_fundamentals(symbol)` — returns a flat dict of fundamental metrics (P&L, Balance Sheet, Cash Flow, ratios). Converts raw Yahoo Finance values to INR crores and percentage formats.

[app/data/ingestion.py](#)

Orchestrates the full data pipeline:

- Upserts stocks, prices, and fundamentals into PostgreSQL
- Runs for all symbols in `UNIVERSE`
- Safe to re-run — uses `ON CONFLICT DO UPDATE` to avoid duplicates

[app/models/](#)

Three SQLAlchemy ORM models:

- `Stock` — ticker symbol, name, sector, industry
- `StockPrice` — daily OHLCV + adjusted close, indexed by `(stock_id, date)`
- `StockFundamentals` — full fundamental snapshot per stock per period, covering valuation, profitability, income statement, balance sheet, cash flow, and growth metrics

[app/backtest/filters.py](#)

`apply_filters(df, filters)` — filters the stock universe by:

- Market cap range (`market_cap_min` , `market_cap_max`)
- `roce_min` , `roe_min` , `pe_max` , `pat_min`

Filters are applied at every rebalance using the latest available fundamental snapshot.

[app/backtest/ranking.py](#)

`rank_stocks(df, ranking, portfolio_size)` — selects the top N stocks by:

- Single metric: direct sort ascending or descending
- Multiple metrics: composite ranking by averaging individual rank positions across all selected metrics

[app/backtest/sizing.py](#)

`compute_weights(df, method, metric)` — computes portfolio weights:

- `equal` — equal weight across all holdings
- `market_cap` — proportional to market capitalisation
- `metric` — proportional to any chosen fundamental metric (e.g. ROCE)

[app/backtest/metrics.py](#)

- `calculate_metrics(equity_curve, initial_capital)` — computes CAGR, Total Return, Sharpe Ratio (risk-free rate 6.5%), Max Drawdown, Volatility, and the full drawdown time series
- `top_performers(portfolio_log)` — identifies top 5 winners and losers by average period return across all rebalances

[app/backtest/engine.py](#)

Core backtest loop:

1. Generates rebalance dates from start to end at chosen frequency
2. Loads all fundamentals and price data from the database
3. At each rebalance: filters → ranks → sizes → computes period return → compounds capital
4. Fetches Nifty 50 (`^NSEI`) equity curve as benchmark via yfinance
5. Returns equity curve, drawdown series, portfolio log, metrics, winners/losers, and benchmark

Note: yfinance only provides current fundamental snapshots, not historical ones. The same latest snapshot is used across all rebalance periods. This is a known limitation of the free data source.

[app/routes/backtest.py](#)

`POST /api/backtest/run` — accepts a JSON config and returns the full backtest result. Required fields: `start_date` , `end_date` .

[app/routes/data.py](#)

Utility endpoints for data status and stock listings.

[app/routes/health.py](#)

`GET /health` — health check endpoint used by Render.

Frontend Modules

[src/api.ts](#)

Single `runBacktest(config)` function that posts to the backend and returns typed results. Backend URL is read from the `VITE_API_URL` environment variable.

[src/types.ts](#)

TypeScript interfaces for `BacktestConfig`, `BacktestResult`, `RebalancePeriod`, `Holding`, `Performer`, and `RankingRule`.

[src/components/ConfigPanel.tsx](#)

Left-panel form with all strategy parameters:

- Date range, rebalance frequency, portfolio size, initial capital
- Position sizing method (and metric selector for metric-weighted)
- Collapsible filters section (market cap, ROCE, ROE, PE, PAT)
- Dynamic ranking rules builder with single/composite support

[src/components/Results.tsx](#)

Container for all output sections. Includes the CSV export button.

[src/components/SummaryCards.tsx](#)

Displays CAGR, Total Return, Sharpe Ratio, Max Drawdown, Volatility, and capital summary as metric cards.

[src/components/EquityChart.tsx](#)

Line chart comparing the strategy equity curve against Nifty 50 benchmark over the backtest period.

[src/components/DrawdownChart.tsx](#)

Area chart showing the portfolio drawdown percentage over time.

[src/components/WinnersLosers.tsx](#)

Two tables showing top 5 best and worst performing stocks by average period return.

[src/components/PortfolioLog.tsx](#)

Paginated accordion log of every rebalance period — date, period return, capital after rebalance. Expandable rows show each holding's weight and return for that period.

[src/lib/exportCsv.ts](#)

Exports the full backtest result (metrics + equity curve + portfolio log) as a CSV file download.

[src/lib/chartDefaults.ts](#)

Registers Chart.js components and sets global chart defaults.

[src/hooks/useTheme.ts](#)

Manages dark/light mode toggle, persisted in `localStorage`.

API Reference

POST /api/backtest/run

Request body:

```
{
  "start_date": "2020-01-01",
  "end_date": "2024-01-01",
  "rebalance_frequency": "quarterly",
  "portfolio_size": 20,
  "initial_capital": 1000000,
  "sizing_method": "equal",
  "sizing_metric": null,
  "filters": {
    "market_cap_min": 1000,
    "roce_min": 15,
    "pat_min": 0
  },
  "ranking": [
    { "metric": "roe", "ascending": false },
    { "metric": "pe_ratio", "ascending": true }
  ]
}
```

Response: equity curve, benchmark curve, portfolio log, metrics (CAGR, Sharpe, drawdown, volatility), top winners and losers.

Optional Features Implemented

- **Nifty 50 benchmark comparison** — equity curve overlay on the chart
 - **Flask REST API** — fully usable via curl or any HTTP client independently of the frontend
 - **CSV export** — download full results including metrics, equity curve, and per-period holdings
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Assumptions

- Fundamental data is sourced from Yahoo Finance's current snapshot. Historical point-in-time fundamentals are not available via the free API, so the same latest snapshot is used at every rebalance date. In a production system this would be replaced with a paid data provider.
- Prices use adjusted close to account for splits and dividends.
- The risk-free rate for Sharpe Ratio calculation is set to 6.5% (approximate Indian T-bill rate).
- All monetary values are in INR crores unless otherwise stated.
- The stock universe is fixed at 120 NSE-listed companies defined in `universe.py`.